



TRIOS CYBER

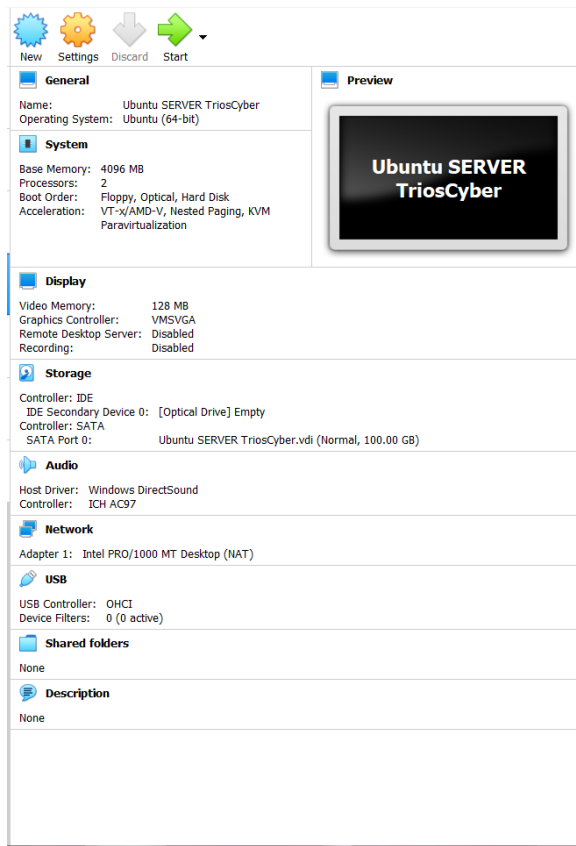
LABORATORY PRACTICE 16

Wazuh Server VM Deployment Preparation and Network Planning

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1. Wazuh Server Virtual Machine Preparation

A dedicated Ubuntu Server virtual machine was prepared in Oracle VirtualBox as the target system for the Wazuh server deployment. The virtual machine was configured with Ubuntu (64-bit), 4096 MB of memory, 2 virtual processors, a 100 GB virtual disk, and an Intel PRO/1000 MT Desktop network adapter attached to NAT. The server started successfully and provided console access for the pre-installation checks.



```

Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of vie 18 sep 2026 02:47:14 UTC

System load: 0.42               Processes:      119
Usage of /: 28.6% of 5.75GB     Users logged in: 0
Memory usage: 4%               IPv4 address for enp0s3: 10.0.2.15
Swap usage: 0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.
   https://ubuntu.com/workshop

El mantenimiento de seguridad expandido para Applications está desactivado.
Se pueden aplicar 51 actualizaciones de forma inmediata.
1 de estas es una actualización de seguridad estándar.
Para ver estas actualizaciones adicionales, ejecute: apt list --upgradable

Active ESM Apps para recibir futuras actualizaciones de seguridad adicionales.
Vea https://ubuntu.com/esm o ejecute «sudo pro status»

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

matthew@trioscyber:~$ hostnamectl
Host: trioscyber
Icon name: computer-vm
Chassis: vm
Machine ID: e8b46e88e9ad7de8e561ccf21365d98
Root ID: c9f1e6938af361bfdb8eb4bdf246f
Virtualization: oracle
Operating System: Ubuntu 24.04.4 LTS
Kernel: Linux 7.0.0-38-generic
Architecture: x86_64
Hardware Vendor: Innovek GmbH
Hardware Model: VirtualBox
Firmware version: VirtualBox
Firmware date: Fri 2006-12-01
Firmware size: 17y 5month 2w 3d
matthew@trioscyber:~$
  
```

2. Supported Operating System and Hostname Verification

The operating system and processor architecture were verified from the server console before any Wazuh installation was performed. The system reported Ubuntu 24.04.4 LTS (Noble Numbat) with x86_64 architecture and the hostname trioscyber. Ubuntu 24.04 is listed as a supported operating system for the Wazuh server, and the 64-bit architecture satisfies the platform requirement.

```

matthew@trioscyber:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
matthew@trioscyber:~$ hostname
trioscyber
matthew@trioscyber:~$ uname -m
x86_64
matthew@trioscyber:~$
  
```

3. IP Configuration and Connectivity Verification

Network configuration was reviewed with `ip -br a` and `ip route`. The active interface `enp0s3` obtained `10.0.2.15/24` through the VirtualBox NAT network, and the default route used gateway `10.0.2.2`. External reachability was tested with four ICMP echo requests to `8.8.8.8`. All four replies were received with 0% packet loss, confirming that the server had working IP connectivity before installation.

```
matthew@trioscyber:~$ ip -br a
lo                UNKNOWN      127.0.0.1/8 ::1/128
enp0s3            UP              10.0.2.15/24 metric 100 fe80::a00:27ff:fe3d:82a6/64
matthew@trioscyber:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev enp0s3 proto dhcp scope link src 10.0.2.15 metric 100
192.168.100.1 via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
matthew@trioscyber:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=113 time=30.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=113 time=71.0 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=113 time=26.8 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=113 time=22.4 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3080ms
rtt min/avg/max/mdev = 22.387/37.619/70.980/19.464 ms
matthew@trioscyber:~$
```

4. RAM and CPU Availability

Available memory and processor capacity were checked with `free -h` and `nproc`. The guest reported approximately 3.8 GiB of RAM with about 3.4 GiB available at the time of the check, and `nproc` reported 2 virtual CPU cores. These values meet the minimum Wazuh server requirement of 2 GB RAM and 2 CPU cores; the configured memory is also close to the 4 GB server recommendation.

```
matthew@trioscyber:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3,8Gi        388Mi        3,3Gi         1,0Mi        337Mi        3,4Gi
Swap:           0B           0B           0B
matthew@trioscyber:~$ nproc
2
```

Resource	Observed	Wazuh server reference	Assessment
RAM	4096 MB configured / ~3.8 GiB detected	Minimum 2 GB	Ready
CPU	2 vCPU	Minimum 2 cores	Ready - minimum

5. Disk Availability Review

Disk resources were reviewed with `lsblk` and `df -h /`. VirtualBox provides a 100 GB virtual disk. Inside the guest, the current root logical volume is approximately 9.8 GB, with about 6.5 GB available during the check. This confirms usable storage for the preparation activity, while also showing that the guest is not currently using the full virtual-disk capacity. For a larger or long-term Wazuh deployment, the root/LVM allocation should be expanded so additional disk capacity is available for collected security data.

```
matthew@trioscyber:~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0    0  100G  0 disk
├─sda1                              8:1    0    1M  0 part
├─sda2                              8:2    0   1,8G  0 part /boot
├─sda3                              8:3    0  18,2G  0 part
└─ubuntu--vg-ubuntu--lv 252:0    0    10G  0 lvm  /
sr0                                  11:0    1 1024M  1 rom
matthew@trioscyber:~$ df -h /
Filesystem                        Size  Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv 9,8G  2,8G  6,5G  31% /
matthew@trioscyber:~$
```

6. Administrative Access Verification

Administrative access was verified using `whoami`, `sudo -v`, and `sudo -l`. The active account was `matthew`, `sudo` authentication succeeded, and the privilege listing showed that the user may execute all commands through `sudo`. This satisfies the administrative-access requirement because Wazuh server installation and configuration require root-level privileges.

```
matthew@trioscyber:~$ whoami
matthew
matthew@trioscyber:~$ sudo -v
[sudo] password for matthew:
matthew@trioscyber:~$ sudo -l
Matching Defaults entries for matthew on trioscyber:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin, u
User matthew may run the following commands on trioscyber:
    (ALL : ALL) ALL
matthew@trioscyber:~$
```

7. Pre-Installation Readiness Assessment

Check	Observed result	Status
Supported OS / architecture	Ubuntu 24.04.4 LTS, x86_64	Ready
Hostname	trioscyber	Ready
IP connectivity	10.0.2.15/24 via NAT; 8.8.8.8 reachable	Ready
RAM	~3.8 GiB detected	Ready
CPU	2 virtual cores	Ready - minimum
Disk	100 GB virtual disk; ~6.5 GB free on current root LV	Review for growth
Administrative access	matthew with sudo ALL privileges	Ready

8. Final Network Plan Before Wazuh Installation

The final network plan was recorded before installation so the expected addressing, routing, and administrative context are clear. The current laboratory deployment uses the VirtualBox NAT network for server connectivity. This provides outbound access required for package retrieval while keeping the preparation environment simple and controlled.

Network parameter	Final plan
Virtual machine role	Wazuh Server
VirtualBox network mode	Adapter 1 - NAT
Server interface	enp0s3
IPv4 address	10.0.2.15/24 (DHCP)
Default gateway	10.0.2.2
Connectivity test	8.8.8.8 reachable; 0% packet loss
Hostname	trioscyber
Administrative account	matthew with sudo privileges
Installation state	Wazuh not yet installed - preparation completed first

9. Preparation Summary

The required pre-installation checks were completed successfully. The server runs a supported Ubuntu release on a 64-bit architecture, the hostname was identified, Internet/IP connectivity was confirmed, RAM and CPU resources were measured, disk availability was reviewed, and administrative sudo access was validated. The current configuration is suitable for continuing to the Wazuh installation phase in a small laboratory environment, with disk allocation identified as the main resource area to review if the deployment grows.

10. Conclusions

The Wazuh server virtual machine was successfully prepared and documented before installation. Ubuntu Server 24.04.4 LTS x86_64 was verified on Oracle VirtualBox, the trioscyber host had working NAT connectivity at 10.0.2.15/24, and the matthew account had the administrative privileges required for deployment. The resource checks confirmed 4 GB of configured memory, 2 virtual CPU cores, and a 100 GB virtual disk, while the root filesystem currently exposes approximately 9.8 GB. The final network plan and readiness assessment establish a clear baseline for the next stage of the Wazuh deployment.